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Reimagining Data Center Power: High-Density Architectures for Next-Gen AI Compute

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Outline

- Motivation for higher current densities VRM solution for next gen AI accelerator
- Need for Ecosystem Advancement
- Vertical Power Delivery Requirements
- Call to Action and Next steps

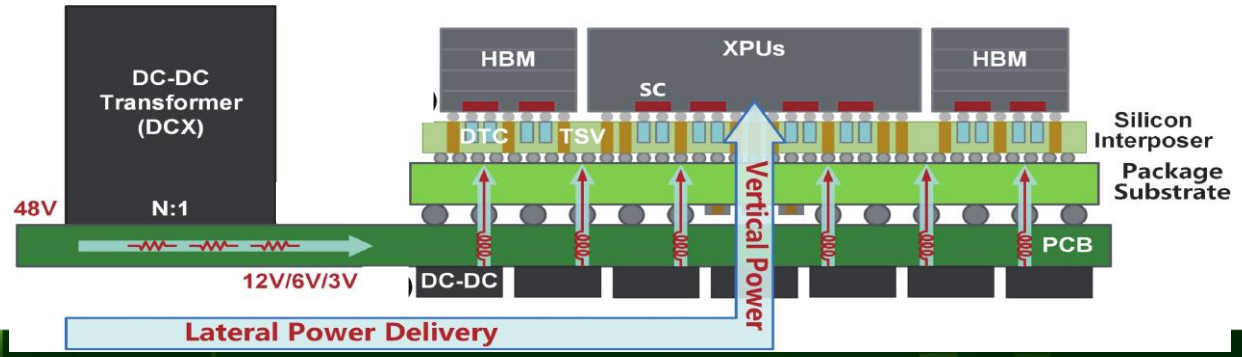
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Accelerator VRM Power Density Advancements

- 50% increase in power density Y2Y over for two consecutive generations drives fundamental changes in power delivery design
  - High-density chiplet integration demands more efficient, higher-density power delivery architecture
  - OAM form factor has reached its limits
  - Ecosystem advancement was required to meet next-generation accelerator power demands
- 2-stage Power Delivery Architecture Advancements
  - 1st Stage - Intermediate Bus Convertor (8:1 IBC)
    - 44V-59.5V input → 5.5V – 7.4V output/ 6.5V nom.
  - 2nd stage multiphase VPD Power modules
- Higher power module density unlocks advanced packaging and thermal techniques previously impractical
  - Backside and vertical power delivery
  - Dual Side Cooling – Bottom side for VPDs, Top Side for GPU + IBC
- Vertical Power Delivery for the best E2E Efficiency

	MI300X -2023	MI355X - 2025	MI455X - 2026
Form Factor	OAM	OAM	EAM
Server Cooling	AC	AC/DLC	DLC
1st Stage IBC	4:1	4:1	8:1
2nd Stage Power Modules(CFP)	Power Module + LPD	Power Module + LPD	Power Module + LPD + VPD
Current Density (A/mm2)	0.88	1.2	Up to 2
Power Density (W/mm3)	0.11	0.15	0.33

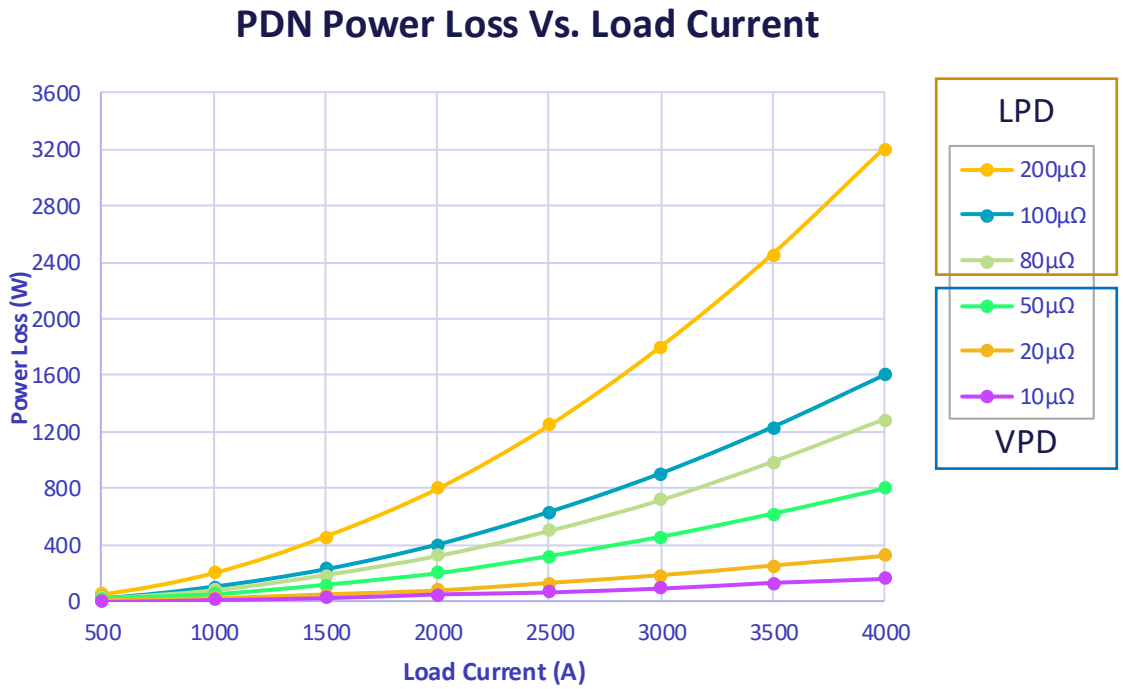
Source: Y Lu, Z G Tong, J C Yang, Z W Yu, M Huang, and X Y Mao, Multi-chip multi-phase DC–DC converters for AI power: a ring, a chain, or a net, independent or master-slave?[J]. J. Semicond., 2025, 46(7), 070201 doi: 10.1088/1674-4926/25040033



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Why Vertical Power ?

- VPD greatly reduces I2R Losses at scale
  - <50 uΩ PDN R-path enables >2 kA w/ >80 % E2E Eff
  - w/o vertical power GPU is power limited
  - PDN is real bottle neck for next gen AI GPUs
- Benefits of VPD
  - Low PDN R+L enables tight regulation and fast transient response
  - Simplifies PCB Power Plane floor planning for high rail count AI GPU cards
  - HSIO and power planes no longer competing for real-estate
  - Low profile VPD power modules enable dual side cooling allowing high current VRMs to be cooled independently of the GPU
  - < 1C/W Enables efficient heat-transfer for VRMs, thus bringing down their overall temp



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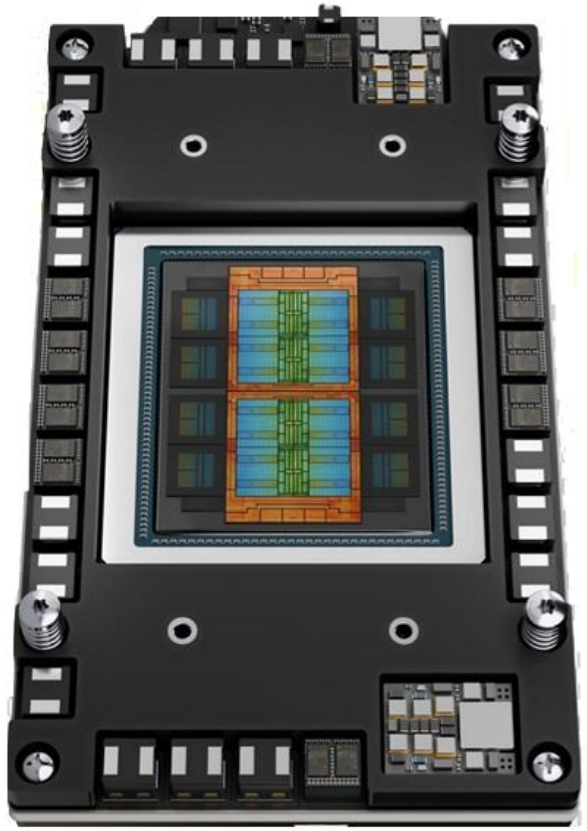
VPD Requirements for Next Gen AI GPUs

	Requirements	Areas of focus
Power Density	❑ High TDC 3A/mm^2 ❑ Low Profile <4mm ❑ Low thermal resistance < 1° C/W	❖ Topology ❖ Inductor ❖ Capacitors ❖ Heat removal innovation ❖ Semiconductor material ❖ Scalability
Efficiency	❑ High efficiency > 80%	❖ E2E design/architecture ❖ PCB PDN Technologies ❖ Semiconductor
Performance	❑ Fast Transient ❑ Tight regulation	❖ E2E design/architecture ❖ Advanced control technique
Reliability	❑ Extra low DPPM ❑ Retire risk early	❖ Improve the design process ❖ Improve ELFR + Advanced telemetry ❖ Comprehensive manufacturing quality

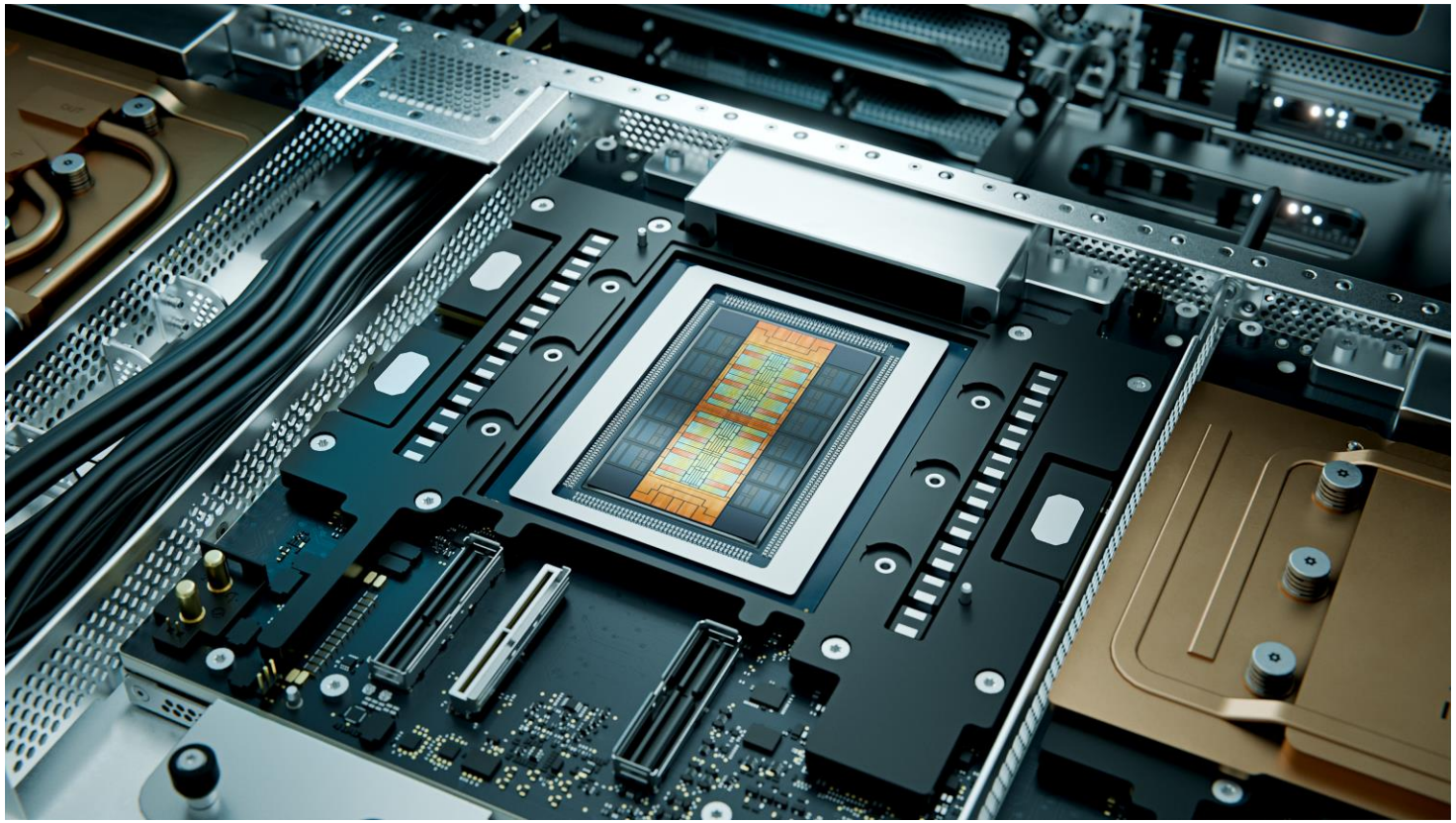
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Platform PD Advancements - MI355X to MI455X

MI355X OAM Top Side – 4:1 IBC + Lateral Power Delivery



MI455X EAM Top Side 8:1 IBC + Lateral Power Delivery



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Power Module Vendor Ecosystem

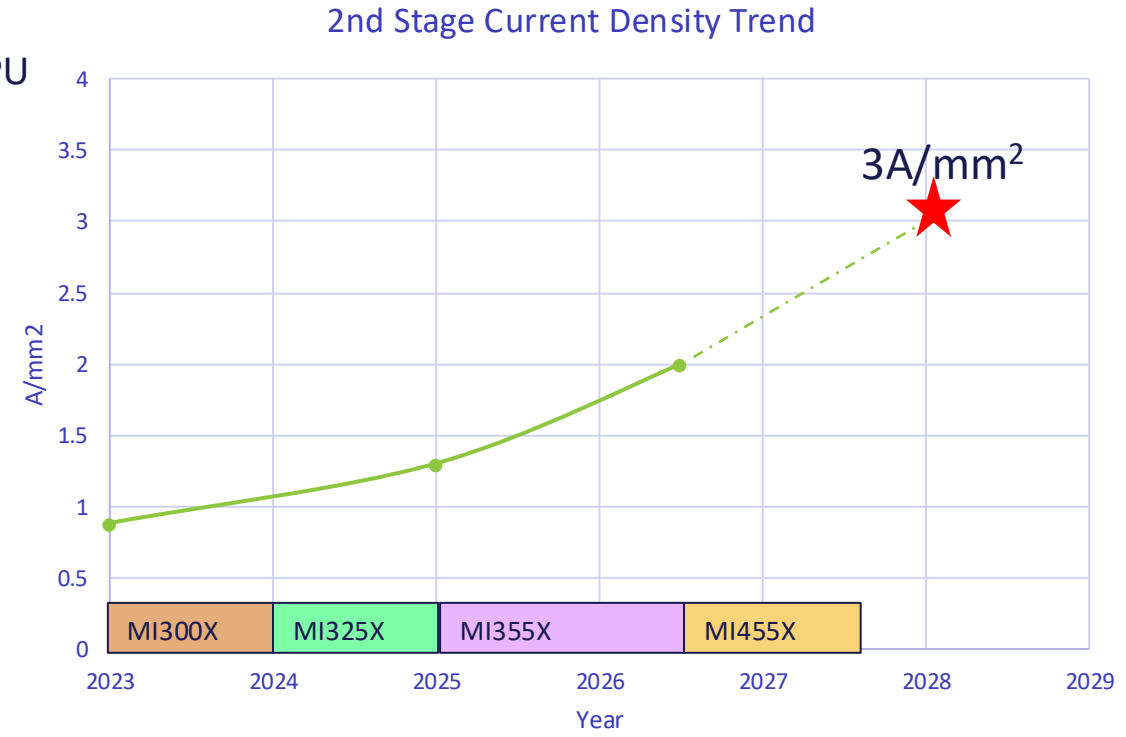
- New technologies enabled by deep technical collaboration and partnership across the ecosystem
  - Concept to production – co-design is a must for success
  - Multi-vendor enabled through standardization of Common-Foot-Print (CFP) and key specs
- Annual cadence of AI accelerators requires faster innovation
  - VRM – Power Blocks + SPS
  - IBCs
  - VR Controllers – Multiphase + multiloop
  - Peripheral – POLs, Efuses
  - Passives – MLCCs, High Current Inductors/magnetic technologies
  - HDI PCB Technologies for PDN Improvements
- Quality and Reliability are fundamental design imperatives
  - At MW scale, DPPM matters

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Call to Action

- MW scale racks become reality within next few years
  - A/mm^2 (W/mm^3) need a paradigm shift to improve E2E eff. Of AI GPU Platforms
  - Vertical Power Delivery becomes standard choice for >2 kW accelerators
  - 3 A/mm^2 + low profile (<4mm) + high Efficiency (>80% E2E) + low thermal resistance (<1C.W) is the goal
- Standardize Vertical Power Module Form Factors and Footprints
- Drive QnR standards for Vertical Power Modules and IBCs
- Pave the way for on package regulators with existing eco-system
- Where to find more information: https://docs.amd.com/



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Thank You!

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